ECE- 600 - Applied Robotics Lab
Checkpoint 9- Autonomous Line Tracking
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Intro:

In this checkpoint, we implemented autonomous line tracking using the robot's onboard camera system. The objective was to enable the robot to detect a colored line on the ground and follow it without manual control. The robot uses its RGB camera to capture images of the environment and processes these images to identify the specified line color. The image is converted into HSV color space to isolate the color of the line more effectively, allowing the system to distinguish the line from the surrounding floor.

Once the line is detected, the program calculates the position of the line relative to the center of the camera frame. This positional difference is used to determine how the robot should steer. If the line appears to the left or right of the center, the robot adjusts its angular velocity to correct its direction while maintaining forward motion. The robot continuously repeats this process as it moves, allowing it to track and follow the line autonomously.

During this checkpoint, we also learned how to configure the camera interface, tune color detection parameters, and integrate the vision-based tracking system with the robot's motion control commands. This exercise demonstrates how computer vision and feedback control can be combined to guide an autonomous mobile robot using visual cues from its environment.

Steps
Legend

T1	Terminal 1
T2	Terminal 2
T3	Terminal 3
T4	Terminal 4

T1

```
roscore
```

T2

```
source ~/catkin_ws/devel/setup.bash
roslaunch yahboomcar_bringup bringup.launch
```

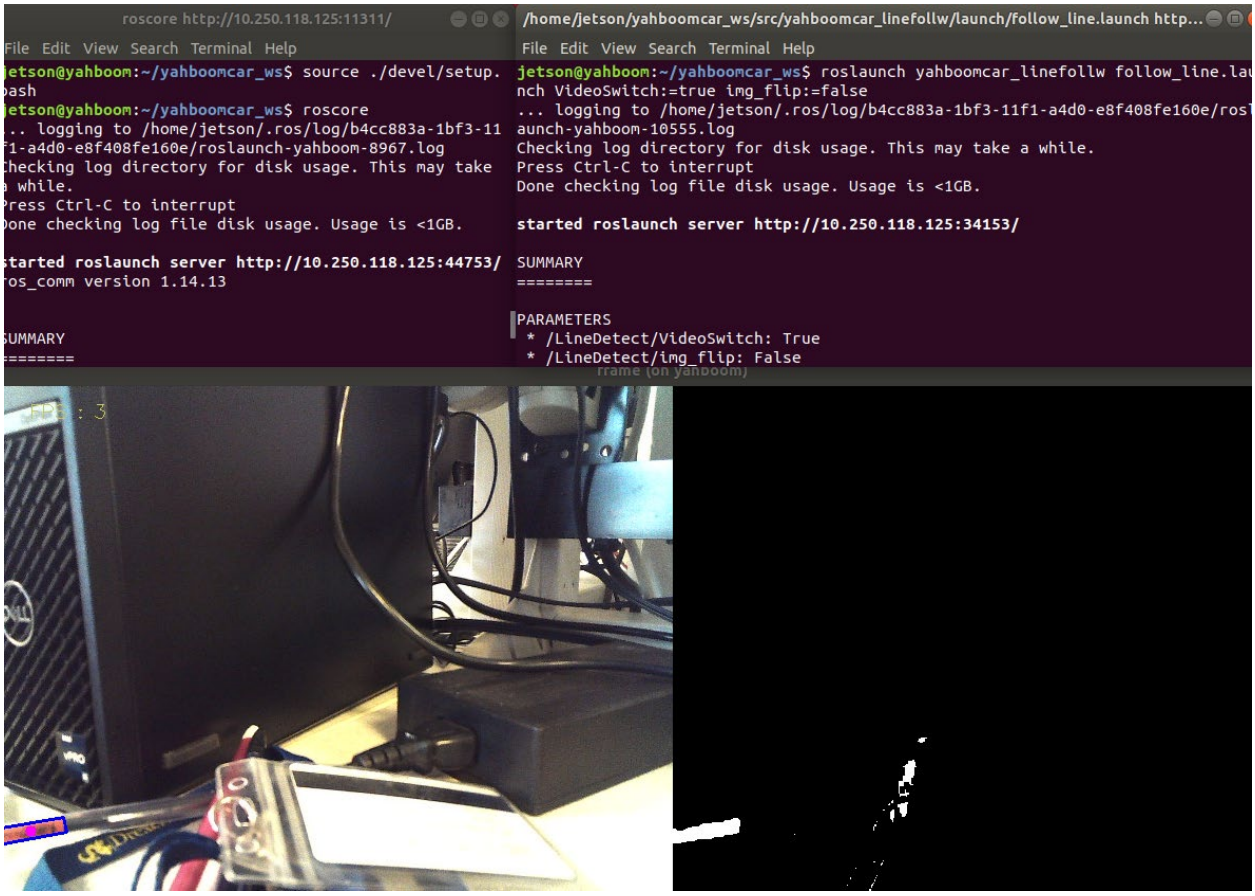
T3

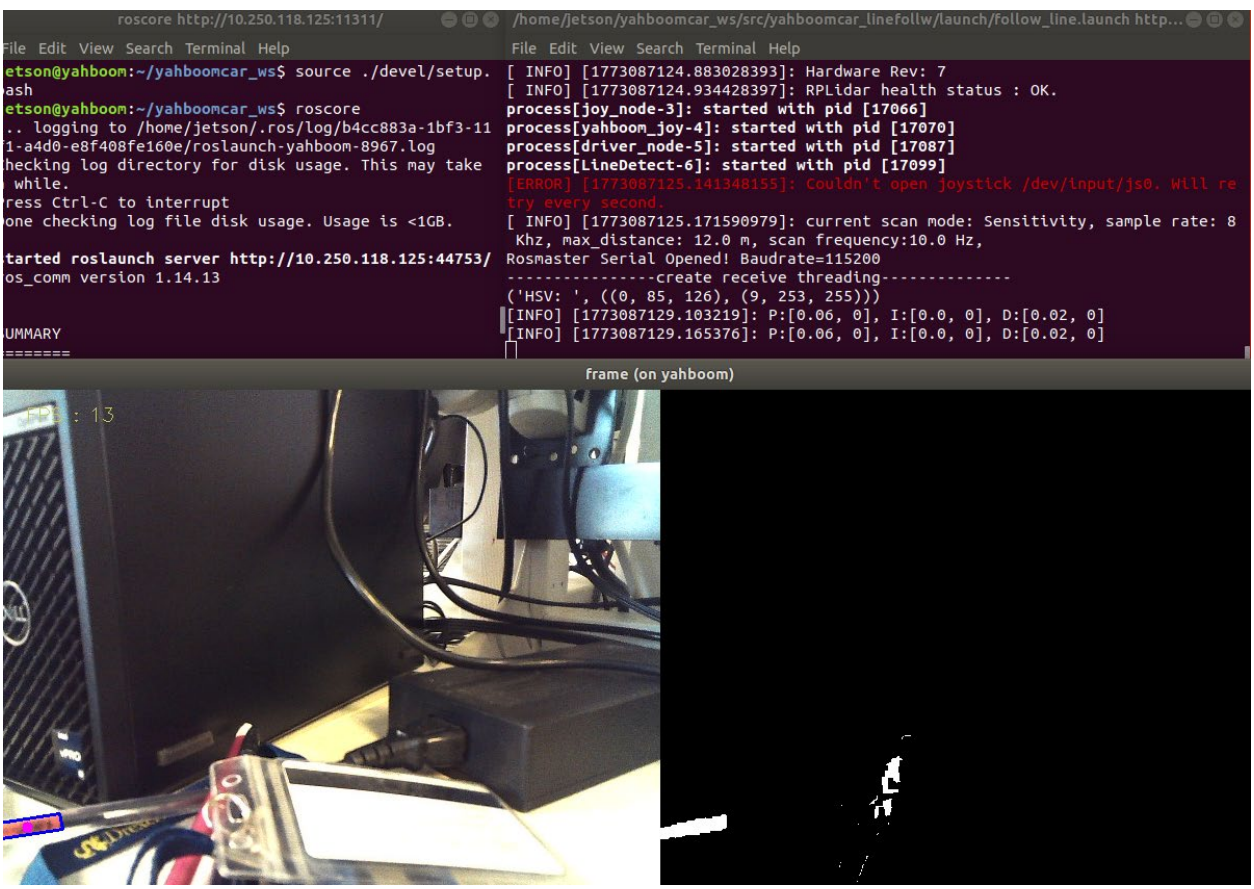
```
cd ~/catkin_ws/src
find . -name follow_line.py
cd ~/catkin_ws/src/yahboomcar_linefollow/scripts
nano follow_line.py
chmod +x follow_line.py
source ~/catkin_ws/devel/setup.bash
roslaunch yahboomcar_linefollow follow_line.launch VideoSwitch:=true img_flip:=false
```

In the window of the camera.

- ‘R’ enters color selection mode
- ‘I’ enters target detection mode
- ‘Space’ starts following
- ‘Q’ - stops the robot

‘I’ is pressed and a click and holding to select the color is done


roscore http://10.250.118.125:11311/
File Edit View Search Terminal Help
etson@yahboom:~/yahboomcar_ws$ source ./devel/setup.
ash
etson@yahboom:~/yahboomcar_ws$ roscore
.. logging to /home/jetson/.ros/log/b4cc883a-1bf3-11
1-a4d0-e8f408fe160e/roslaunch-yahboom-8967.log
hecking log directory for disk usage. This may take
while.
ress Ctrl-C to interrupt
one checking log file disk usage. Usage is <1GB.
tarted roslaunch server http://10.250.118.125:44753/
os_comm version 1.14.13
UMMARY
/home/jetson/yahboomcar_ws/src/yahboomcar_linefollw/launch/follow_line.launch http...
File Edit View Search Terminal Help
jetson@yahboom:~/yahboomcar_ws$ roslaunch yahboomcar_linefollw follow_line.lau
nch VideoSwitch:=true img_flip:=false
... logging to /home/jetson/.ros/log/b4cc883a-1bf3-11f1-a4d0-e8f408fe160e/ros
aunch-yahboom-10555.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.
started roslaunch server http://10.250.118.125:34153/
SUMMARY
PARAMETERS
* /LineDetect/VideoSwitch: True
* /LineDetect/img_flip: False
frame (on yahboom)

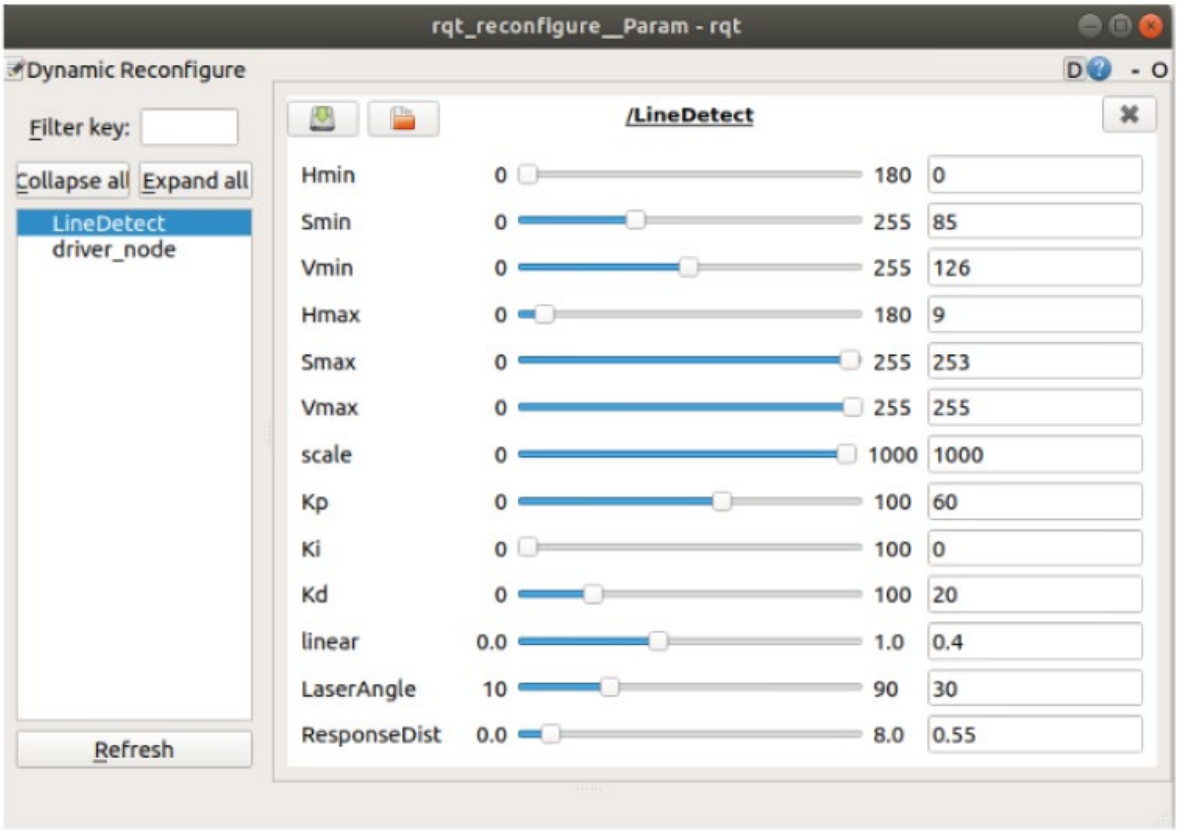

roscore http://10.250.118.125:11311/
File Edit View Search Terminal Help
etson@yahboom:~/yahboomcar_ws$ source ./devel/setup.
ash
etson@yahboom:~/yahboomcar_ws$ roscore
.. logging to /home/jetson/.ros/log/b4cc883a-1bf3-11
1-a4d0-e8f408fe160e/roslaunch-yahboom-8967.log
hecking log directory for disk usage. This may take
while.
ress Ctrl-C to interrupt
one checking log file disk usage. Usage is <1GB.
tarted roslaunch server http://10.250.118.125:44753/
os_comm version 1.14.13
UMMARY
/home/jetson/yahboomcar_ws/src/yahboomcar_linefollw/launch/follow_line.launch http...
File Edit View Search Terminal Help
[INFO] [1773087124.883028393]: Hardware Rev: 7
[INFO] [1773087124.934428397]: RPLidar health status : OK.
process[joy_node-3]: started with pid [17066]
process[yahboom_joy-4]: started with pid [17070]
process[driver_node-5]: started with pid [17087]
process[LineDetect-6]: started with pid [17099]
[ERROR] [1773087125.141348155]: Couldn't open joystick /dev/input/js0. Will re
try every second.
[INFO] [1773087125.171590979]: current scan mode: Sensitivity, sample rate: 8
Khz, max_distance: 12.0 m, scan frequency:10.0 Hz,
Rosmaster Serial Opened! Baudrate=115200
----------create receive threading----------
('HSV: ', ((0, 85, 126), (9, 253, 255)))
[INFO] [1773087129.103219]: P:[0.06, 0], I:[0.0, 0], D:[0.02, 0]
[INFO] [1773087129.165376]: P:[0.06, 0], I:[0.0, 0], D:[0.02, 0]
frame (on yahboom)

Step 8 - tune the HSV parameters

T4

```
rosrun rqt_reconfigure rqt_reconfigure
```

Using RQT configure to adjust the robots response to tracking



Parameter	What It Controls	Basic Purpose
Hmin	Minimum Hue value	Lower bound of the color range used to detect the line. Determines where color detection begins.

Smin	Minimum Saturation	Filters out dull or gray colors so the robot focuses on stronger colored regions.
Vmin	Minimum Brightness (Value)	Ignores very dark pixels such as shadows.
Hmax	Maximum Hue value	Upper bound of the color range used for line detection.
Smax	Maximum Saturation	Limits how intense a color can be while still being detected. Usually left near 255.
Vmax	Maximum Brightness	Prevents overly bright reflections or glare from being detected as the line.
scale	Image scaling factor	Adjusts the size of the processed image to balance speed vs detection detail.
Kp	Proportional gain	Determines how strongly the robot turns when the detected line is off center.
Ki	Integral gain	Corrects long-term steering drift by accumulating error over time.
Kd	Derivative gain	Helps smooth steering by reacting to how quickly the error changes.
linear	Forward speed	Sets the robot's base forward velocity while following the line.
LaserAngle	LiDAR detection angle	Defines the angular range used for obstacle detection during line following.
ResponseDist	Obstacle response distance	Determines how close an obstacle must be before the robot reacts or slows down.

In this case we adjusted the Kp from a value of 60 to 90

Step 9

By confirming all of this, the laptop operator simply needs to active the camera screen and hit ‘space’ bar to active the robot into following the line

The script of ‘follw_line.py’ script processes images from the robot’s RGB camera to identify a colored line using HSV-based color segmentation. Once the line is detected, the script computes the horizontal offset between the center of the line and the center of the image. This offset is treated as an error signal and is used to adjust the robot’s angular velocity while maintaining forward motion. In this way, the robot continuously corrects its steering and follows the line autonomously.

Attached is a quick 30 second video of the robot following the line.